Homework 3: Using Spatial Lag, Spatial Error and Geographically Weighted Regression to Predict Gini Index/income inequality in New York City

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1) Introduction

In the modern United States and much of the developed world, income inequality has become a serious issue as the gap in real wages between the wealthiest and poorest of society continues to grow. Income inequality is nowadays well understood to be the source of significant political frustration, and is harming the wellbeing of our societies globally. In cities such as New York with very high populations, this is a particularly common issue. In this study, we model the Gini index, a measure of inequality, to examine what factors predict inequality and how these relationships change depending on proximity. In particular, we examine the results of spatial lag, spatial error and geographically weighted regression (GWR) to see whether these models can accurately account for the spatial autocorrelation that may occur in OLS residuals.

The key issue is that OLS regression assumes independence among observations in data, an assumption that is likely to be violated in spatial data as neighbouring regions share similar traits and values. In this study, we use three different spatial models to see which is best able to account for the problems with spatial data in comparison to OLS.

2) Methods

(a) Concept of Spatial Autocorrelation

(i) The First Law of Geography is the basic premise in spatial analysis, stating: "Everything is related to everything else, but near things are more related than distant things." Essentially, places nearby to each other tend to be similar, as well as the data relating to them. In this case for our example, regions with high income inequality are generally expected to be near others with high inequality also. This is the concept behind spatial autocorrelation, which shows us the correlation amongst values of a single observation that can be attributed to close proximity to another observation. It is very important to identify this, as the vast majority of statistical techniques assume observations are independent. If spatial autocorrelation is present, those techniques may give us false or misleading results.

(ii) One way to measure spatial autocorrelation is Moran's I Formula, a rather old way of measuring whether similar values (such as Gini indices) are clustered, dispersed, or randomly distributed in space. The formula for this is:

$$= \frac{n}{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}} \frac{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}\,(X_i - \bar{X})(X_j - \bar{X})}{\sum_{i=1}^{n}(X_i - \bar{X})^2}$$

Here, n is the number of observations (census regions). X is the mean of variable X (gini index). X_i is the variable value at one location i, and X_j is the variable value at another location j. W_{ij} is the spatial weight between i and j. A value of Moran's I = 1 means values cluster, a value of -1 means they are dispersed, and a value of 0 implies randomness.

(iii) In spatial models, we use a spatial weights matrix in order to identify which regions are considered neighbours. In order to use this, we used the Queen matrix, where regions are considered neighbours if they share a boundary or a corner. This is more flexible than a Rook matrix, which only includes shared boundaries and not corners. The Queen matrix is therefore better suited to examine spatial influence in environments like NYC, where many regions touch diagonally. Statisticians often use more than one weight matrix in order to ensure that the spatial patterns we see are common amongst them, and not just the result of one particular definition of what a "neighbourhood" is.

(iv) We assess the significance of Moran's I using a random permutation inference. Here, the null hypothesis is that the variable (Gini index) is spatially random. The alternative hypothesis is that there is spatial correlation. In order to test this, values are shuffled randomly across locations 999 times. The observed Moran's I is then compared to this distribution. This gives us a Psuedo-P value, which is the chance of getting a value as large as the observed value using samples from this distribution. We would generally say that if the p-value < 0.05, then the difference is statistically significant. This would lead us to reject the null hypothesis, that is, accept that there is spatial autocorrelation.

(v) Whilst Moran's I tells us if spatial autocorrelation exists, it doesn't show us where they occur. However, Local Moran's I, also known as LISA, can. It gives a measure for each region and helps create cluster maps that show the following in the context of our study:

- High-High: High inequality next to high inequality
- Low-Low: Low inequality next to low
- High-Low and Low-High: Spatial outliers

This is also a useful tool for visualizing the spatial structure of your data, such as the location of inequality in NYC.

(b) Review of OLS Regression and Assumptions

(i) OLS is a very common regression method, but comes with the following assumptions:

1. Linearity - The relationship between predictors and the outcome is linear
2. Normality - Errors should be normally distributed
3. Homoscedasticity - There is constant variance of residuals across all observations
4. Independence - Observations should not influence each other
5. No Multicollinearity - Predictors should not be highly correlated

(ii) In spatial analysis, the independence assumption is commonly violated. This can lead to spatial dependence in residuals, making our OLS results unreliable. To test this, we apply Moran's I to the residuals from the OLS model. A significant Moran's I here would suggest that the OLS has been unable to accurately measure spatial relationships in the data, and needs spatial regression methods. We can also measure this by regressing OLS residuals on nearby residuals. The slope of the resulting regression line gives us an estimate of rho (ρ), with a strong positive slope confirming spatial dependence. In the context of spatial regression, ρ captures the extent of spatial autocorrelation in the variable, in this case the gini index. If it is significantly high, it suggests that the value in one region is influenced by the value in nearby regions.

(iii) GeoDa, one of the software used in this analysis, has a way to test the validity of OLS assumptions. The Breusch-Pagan Test checks for heteroscedasticity, tied to the assumption of independence of errors. The Null hypothesis is that residuals have constant variance (homoscedasticity) with the alternative hypothesis being that they have heteroscedasticity. The Koenker-Bassett Test also tests for the same thing, it does not assume normality of residuals. The Jarque-Bera test is used to test for the normality of errors. The Null Hypothesis is that they are normally distributed, with the alternative hypothesis being that they are not.

(c) Spatial Lag and Spatial Error Regression

(i) GeoDa will be used to run spatial lag and spatial error regressions.

(ii) The spatial lag model assumes that the outcome in one region is directly affected by the outcomes in nearby regions. It adds a spatially lagged dependent variable to the regression. This is very useful when it is expected that regions will affect each other, for example if high inequality in one area spreads into another. The formula is:

$$\text{Gini}_i = \rho \text{WGini}_i + \beta_0 + \beta_1 \cdot \text{bachelor}_i + \beta_2 \cdot \text{withpubass}_i + \beta_3 \cdot \text{lessthan10}_i + \beta_4 \cdot \text{households}_i + \beta_5 \cdot \text{femaleunem}_i + \varepsilon_i$$

$\beta 0$ - intercept
ε - Error term
$\rho WGini_i$: - Spatially lagged Gini index

(iii) The spatial error model assumes that the residual at one location is associated with residuals at nearby locations. This can be because of omitted variables that can influence the values and variables but that are not included in the model. The formula is:

$$Gini_i=\beta 0+\beta 1 \cdot bachelor_i+\beta 2 \cdot withpubass_i+\beta 3 \cdot lessthan10_i+\beta 4 \cdot households_i+\beta 5 \cdot femaleunem_i+u_i$$

$$ui=\lambda Wu_i+\varepsilon_i$$

ε - Error term
$\beta 0$ - intercept
$\rho WGini_i$: - Spatially lagged Gini index
λWu_i: Spatial error term

(iv) Spatial regression models relax the OLS independence assumption, but they still assume linearity, normality and there should be no multicollinearity. This is a more appropriate way to model spatial data than OLS when spatial autocorrelation is present.

(v) These models aim to correct bias caused by spatial autocorrelation, improve the accuracy of predictions, show us the spatial influences in our model, and overall make the model more reliable. This will allow us to better understand how inequality spreads or persists in NYC.

(vi) We compare our models in a number of ways. They are:

1. Akaike Information Criterion (AIC) - This assesses the overall fit of the model, balancing both accuracy and complexity. Lower AIC indicates a better model.
2. Log Likelihood - This measures how likely the model is to have generated the observed data. It checks how well the model fits without penalizing complexity. A higher log likelihood means a better fitting model.
3. Likelihood Ratio Test (LRT) - This tests whether the spatial model provides a better fit than OLS. It compares the likelihoods of the two models. The null hypothesis is that the OLS model fits the data just as well and that the spatial model adds no benefit. The alternative hypothesis is that the spatial model significantly improves the fit over OLS. If LRT p-value is < 0.05, we reject the null hypothesis.

Another way to compare models is by checking the Moran's I on their residuals. This shows us how well each model has removed spatial autocorrelation. Lower and less significant Moran's I values on residuals suggest better model performance. If the Moran's I of the residuals is close to 0, the model is deemed to have successfully accounted for spatial issues.

(d) Geographically Weighted Regression (GWR)

(i) GWR allows the relationships between predictors and outcomes to vary depending on location. Instead of using just one single global regression, it runs a local regression for each region, weighting each observation differently.

(ii) Global patterns can mask local variations, which is called Simpson's Paradox. There may be two different relationships between variables in different locations. For example, the effect of financial assistance may have the opposite effect on inequality in one area than another. GWR can help us see these local differences.

(iii) The formula for GWR is:

$$y_i = \beta_{i0} + \beta_{i1}x_{i1} + \beta_{i2}x_{i2} + \cdots + \beta_{im}x_{im} + \varepsilon_i = \beta_{i0} + \sum_{k=1}^{m} \beta_{ik}\, x_{ik} + \varepsilon_i$$

yi - The dependent variable (gini index at location i)
xik - The value of the predictor variable at location i
βik - The regression coefficient for the predictor at location i
Bi0 - The intercept at location i
m - The total number of predictor variables
εi - Error term

(iv) GWR works by using other observations in the dataset to run the regression, as you need multiple observations to run a regression, not just one. Observations that are close to location i are given greater weights. This process is repeated for every location in the dataset, allowing it to model spatially varying relationships.

(v) Bandwidth determines how many neighbours are included based on distance. This can greatly change the results, as the assumptions you make about weights greatly affect them. The best way to deal with this in ArcGIS is to allow it to do it automatically for you, and by choosing the golden search method. It will find the best values for distance band and number of neighbours.

(vi) GWR still holds a number of assumptions. It assumes local linearity, local normality of residuals, and also does not allow severe local multicollinearity. This is checked through the condition number. It measures the stability of a local regression model. When the value for variables cluster spatially in a substantial way, it is likely to cause issues with multicollinearity.

(vii) Due to the fact GWR runs many regressions, testing significance at every single location would be likely to cause false positives because of simply how many tests there are, due to the standard of error. Due to this, p-values are not shown in GWR outputs.

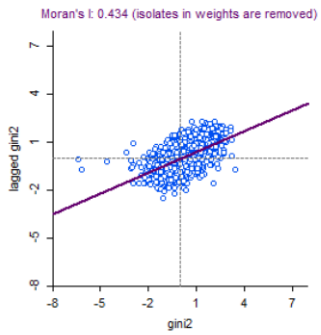
(viii) For the purpose of this report, we will only be evaluating GWR with three global diagnostics that we obtain from GeoDa:

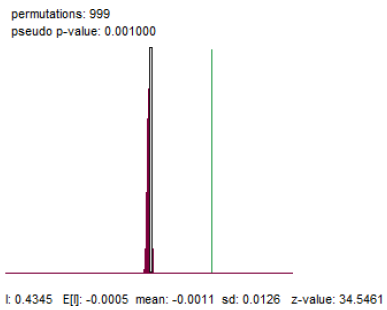
1. Adjusted R-Squared
2. AIC
3. Residual Moran's I

3) Results

(a) Spatial Autocorrelation

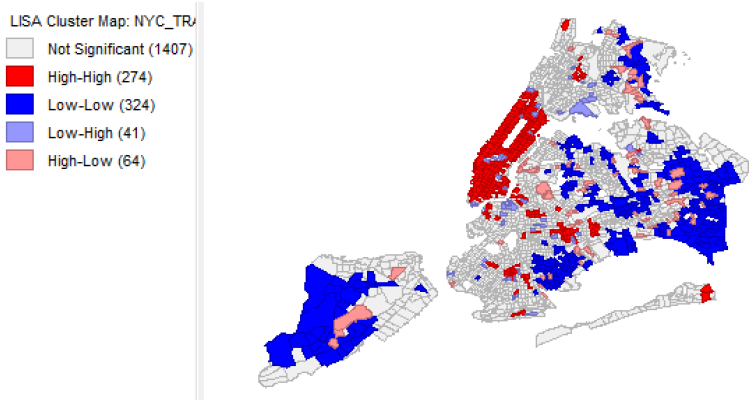
(i) Global Moran's I was used to test whether the Gini index shows spatial clustering across NYC.



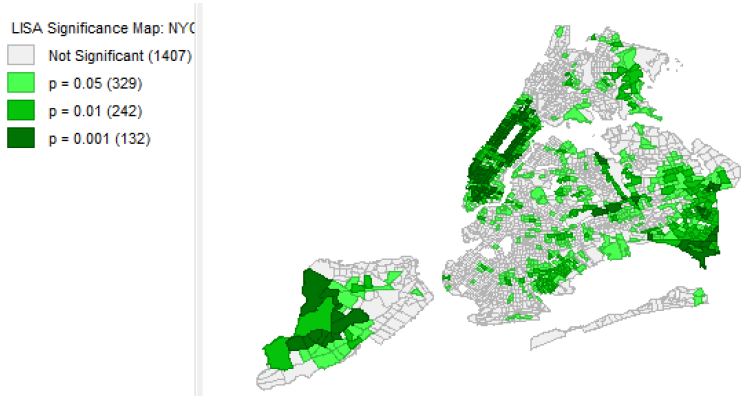


The Moran's I was 0.434, with the z-score being 34.55, and p-value being 0.001. This indicates a clear strong positive spatial autocorrelation in the Gini index. In other words, areas with high inequality tend to be near other high-inequality areas. This violates the assumption of independence in standard regression models.

(ii) The Local Moran's I (LISA) cluster map shows local pockets of spatial clustering.



High-High clusters were observed in Manhattan in particular, with this area having high inequality alongside its neighbours. Low-Low clusters were observed mainly in Staten Island, alongside the eastern sides of Queens. These are areas with low inequality surrounded by others with low inequality. Non-significant areas cover the entire city .



We see the areas with High-High and Low-Low clusters have similarly unique levels of significance.

(b) OLS Regression

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>>07/22/25 23:00:33
REGRESSION
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SUMMARY OF OUTPUT: ORDINARY LEAST SQUARES ESTIMATION
Data set           : NYC_TRACK_ACS_EXPORT
Dependent Variable  :     gini2  Number of Observations: 2112
Mean dependent var  :   0.443775  Number of Variables   :    6
S.D. dependent var  :  0.0668558  Degrees of Freedom    : 2106

R-squared           :   0.306652  F-statistic           :   186.287
Adjusted R-squared  :   0.305006  Prob(F-statistic)     :         0
Sum squared residual:    6.54521  Log likelihood        :   3103.35
Sigma-square        : 0.00310788  Akaike info criterion :  -6194.71
S.E. of regression  : 0.0557484  Schwarz criterion     :  -6160.77
Sigma-square ML     : 0.00309906
S.E of regression ML:  0.0556692

-----------------------------------------------------------------------
     Variable     Coefficient     Std.Error   t-Statistic   Probability
-----------------------------------------------------------------------
     CONSTANT        0.4185     0.00229834       182.088     0.00000
     bachelor      7.21e-05    5.68544e-06       12.6815     0.00000
   withpubass   7.35664e-05    2.89953e-05       2.53719     0.01125
   lessthan10   0.000282612    1.42718e-05       19.8021     0.00000
   households  -3.37995e-05     3.8723e-06      -8.72853     0.00000
   femaleunem  -0.000119374    2.04696e-05      -5.83176     0.00000
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REGRESSION DIAGNOSTICS
MULTICOLLINEARITY CONDITION NUMBER 14.390986
TEST ON NORMALITY OF ERRORS

TEST	DF	VALUE	PROB
Jarque-Bera	2	963.9031	0.00000

DIAGNOSTICS FOR HETEROSKEDASTICITY
RANDOM COEFFICIENTS

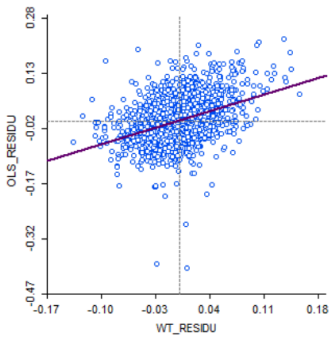
TEST	DF	VALUE	PROB
Breusch-Pagan test	5	61.8008	0.00000
Koenker-Bassett test	5	23.2789	0.00030

DIAGNOSTICS FOR SPATIAL DEPENDENCE
FOR WEIGHT MATRIX : NYC_TRACK_ACS_EXPORT
(row-standardized weights)

TEST	MI/DF	VALUE	PROB
Moran's I (error)	0.2525	19.7006	0.00000
Lagrange Multiplier (lag)	1	468.7911	0.00000
Robust LM (lag)	1	101.4268	0.00000
Lagrange Multiplier (error)	1	381.3142	0.00000
Robust LM (error)	1	13.9500	0.00019
Lagrange Multiplier (SARMA)	2	482.7410	0.00000

Table 1

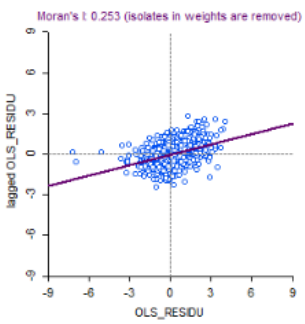
(i) This model explains about 30.7% of the variation in inequality seen in our R-squared value. This is a fairly poor explanatory power, and we should expect better performance from our spatial models. All of our predictors are significant. We can see from our assumption tests that this model fails key assumptions of constant variance and normal errors. They are all consistent with each other, and show substantial problems with heteroscedasticity as seen in the Breusch-Pagan and Koenker-Bassett test. The Jarque-Bera test also shows they are not normally distributed.

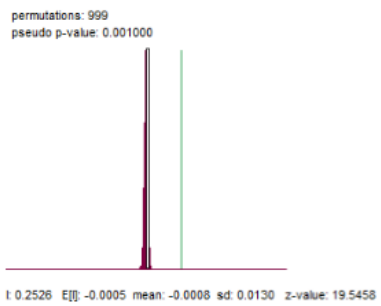


#obs	R^2	const a	std-err a	t-stat a	p-value a	slope b	std-err b	t-stat b	p-value b
2110	0.163	0.001	0.001	1.022	0.307	0.647	0.032	20.287	0
0	0	0	0	0	0	0	0	0	0
2110	0.163	0.001	0.001	1.022	0.307	0.647	0.032	20.287	0

Chow test for sel/unsel regression subsets: need two valid regressions

(ii) We see from the scatterplot of OLS_RESIDU and WT_RESIDU that there is a strong positive spatial autocorrelation in residuals. Areas with large errors tend to be near others with similar areas. 16.3% of the variation in OLS residuals is explained by the residuals of their neighbours.





(iii) Through this scatterplot of the OLS regression residuals, again, we confirm that there is strong spatial autocorrelation, which violates OLS assumptions. The upward sloping line confirms clustering, and this is very problematic for our model. Our observations are not independent.

(c) Spatial Lag Model

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>>07/22/25 23:30:53
REGRESSION
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SUMMARY OF OUTPUT: SPATIAL LAG MODEL - MAXIMUM LIKELIHOOD ESTIMATION
Data set          : NYC_TRACK_ACS_EXPORT
Spatial Weight     : NYC_TRACK_ACS_EXPORTffffffffffffffff
Dependent Variable  :      gini2  Number of Observations: 2112
Mean dependent var  :   0.443775  Number of Variables   :   7
S.D. dependent var  :  0.0668558  Degrees of Freedom    : 2105
Lag coeff.  (Rho)  :   0.424511

R-squared           :   0.431504  Log likelihood        :   3239.33
Sq. Correlation     : -          Akaike info criterion :   -6464.65
Sigma-square        :   0.002541  Schwarz criterion     :   -6425.06
S.E of regression   :   0.0504084

-----------------------------------------------------------------------------
    Variable     Coefficient    Std.Error     z-value   Probability
-----------------------------------------------------------------------------
      W_gini2      0.424511     0.0232755      18.2386     0.00000
     CONSTANT      0.239081     0.0101919      23.4581     0.00000
   withpubass   3.59522e-05    2.6244e-05      1.36992     0.17071
   lessthan10  0.000220854   1.33241e-05      16.5755     0.00000
   femaleunem  -8.53587e-05   1.85288e-05     -4.60682     0.00000
     bachelor   4.61607e-05   5.23524e-06      8.8173     0.00000
   households  -2.40962e-05   3.52922e-06     -6.82763     0.00000
-----------------------------------------------------------------------------
```

Table 2

```
REGRESSION DIAGNOSTICS
DIAGNOSTICS FOR HETEROSKEDASTICITY
RANDOM COEFFICIENTS
TEST                          DF     VALUE      PROB
Breusch-Pagan test             5     29.5033    0.00002

DIAGNOSTICS FOR SPATIAL DEPENDENCE
SPATIAL LAG DEPENDENCE FOR WEIGHT MATRIX : NYC_TRACK_ACS_EXPORTffffffffffffffff
TEST                          DF     VALUE      PROB
Likelihood Ratio Test          1     271.9456   0.00000
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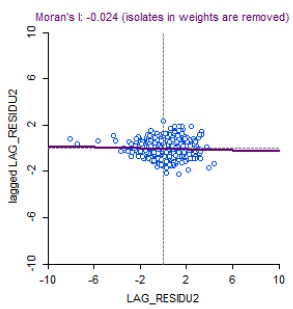
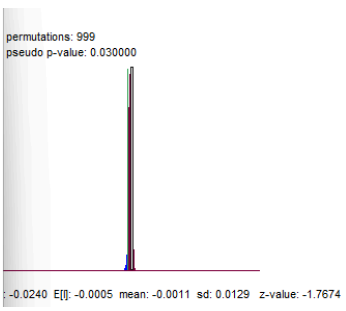
(i) Using the calculator to get the spatial lagged version of gini, there was a huge difference in the values. In cases where WT_Gini2 was larger than gini2, its neighbours have more inequality than it does. Otherwise, it implied its neighbours have less inequality. It shows a strong spillover effect.
Compared to the OLS, most predictors remain significant. However, withpubass variable loses significance in the lag model. Here, the Breusch-Pagan test result shows that there is still evidence of heteroscedasticity with a value of 29.5033 with a p-value of 0.00002. However, we can see that the spatial lag model performs significantly better than OLS on all other criteria. There are lower AIC and Schwarz Criterion values and a stronger LR test. The likelihood ratio test also has a p-value beneath 0.001.





We can see from the Moran's I scatterplot of spatial lag regression residuals that there is far less spatial autocorrelation in these residuals. It shows a very slight downward slope, suggesting weak negative spatial autocorrelation - a major improvement over the OLS residuals with a Moran's I-value of -0.024, z-value of -1.7674, and a p-value of 0.03. Overall, the spatial lag model is clearly better than OLS.

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>>07/22/25 23:32:26
REGRESSION
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SUMMARY OF OUTPUT: SPATIAL ERROR MODEL - MAXIMUM LIKELIHOOD ESTIMATION
Data set           : NYC_TRACK_ACS_EXPORT
Spatial Weight     : NYC_TRACK_ACS_EXPORTffffffffffffffff
Dependent Variable  :     gini2  Number of Observations: 2112
Mean dependent var  :   0.443775  Number of Variables   :    6
S.D. dependent var  :   0.066856  Degrees of Freedom    : 2106
Lag coeff. (Lambda) :   0.522698

R-squared           :   0.433505  R-squared (BUSE)     : -
Sq. Correlation     : -           Log likelihood       : 3223.181202
Sigma-square        :  0.00253206  Akaike info criterion :   -6434.36
S.E of regression   :  0.0503196  Schwarz criterion    :   -6400.43

------------------------------------------------------------------------
    Variable     Coefficient    Std.Error      z-value   Probability
------------------------------------------------------------------------
    CONSTANT      0.428481    0.00306045       140.006     0.00000
  withpubass   3.26029e-05   2.73488e-05       1.19212     0.23322
  lessthan10   0.000234105   1.40587e-05        16.652     0.00000
  femaleunem  -6.07619e-05   1.9832e-05      -3.06384     0.00219
   bachelor   4.21454e-05   5.95511e-06       7.07718     0.00000
  households  -2.56448e-05   3.77549e-06      -6.79244     0.00000
     LAMBDA      0.522698    0.0256137        20.407     0.00000
------------------------------------------------------------------------

REGRESSION DIAGNOSTICS
DIAGNOSTICS FOR HETEROSKEDASTICITY
RANDOM COEFFICIENTS
TEST                          DF      VALUE       PROB
Breusch-Pagan test             5     22.1924     0.00048

DIAGNOSTICS FOR SPATIAL DEPENDENCE
SPATIAL ERROR DEPENDENCE FOR WEIGHT MATRIX : NYC_TRACK_ACS_EXPORTffffffffffffffff
TEST                          DF      VALUE       PROB
Likelihood Ratio Test          1     239.6573     0.00000
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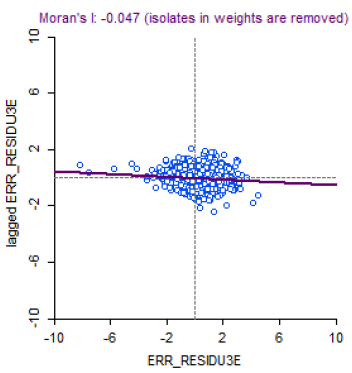
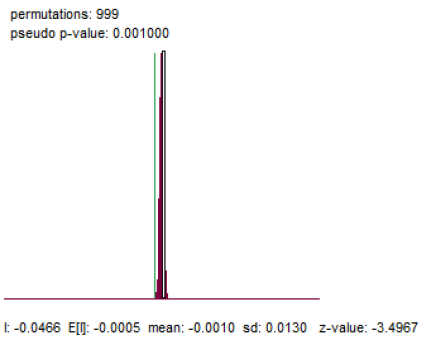
Table 3

(ii) Our LAMBDA coefficient is 0.5227, highly significant. This would suggest that as mentioned previously unmeasured spatial effects are influencing the Gini index, which the spatial error model is accounting for. As with the previous model, withpubass becomes insignificant. Other predictors remain the same. With our Breusch-Pagan test with a score of 22.1924 and a p-value of 0.00048, we see that heteroscedasticity still exists, although slightly reduced compared to the lag model. The spatial error model also clearly outperforms OLS in AIC, Schwarz Criterion, Log Likelihood and the Likelihood Ratio test.





Here, we see that the residuals now show a further reverse in spatial autocorrelation, now of an opposite kind. They are now negatively correlated, suggesting dispersion. Despite this, it is still much improved from the OLS. Overall, it is again superior to OLS.

Altogether however, the lag model is performing slightly better. Its AIC is lower, as is its Schawrz Criterion, and it seems to have balanced spatial autocorrelation better than the error model.

(d) Geographically Weighted Regression (GWR) Results

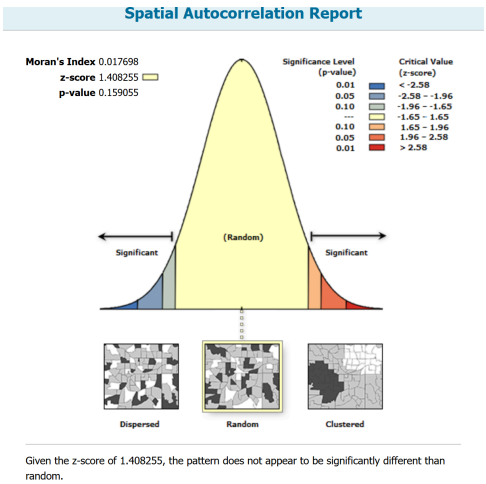
Model Diagnostics

R2	0.5717
AdjR2	0.5095
AICc	-6776.3114
Sigma-Squared	0.0022
Sigma-Squared MLE	0.0019
Effective Degrees of Freedom	1844.5979
Adjusted Critical Value of Pseudo-t Statistics	3.1807

Global Moran's I Summary

Moran's Index	-0.001288
Expected Index	-0.000474
Variance	0.000020
z-score	-0.182279
p-value	0.855364

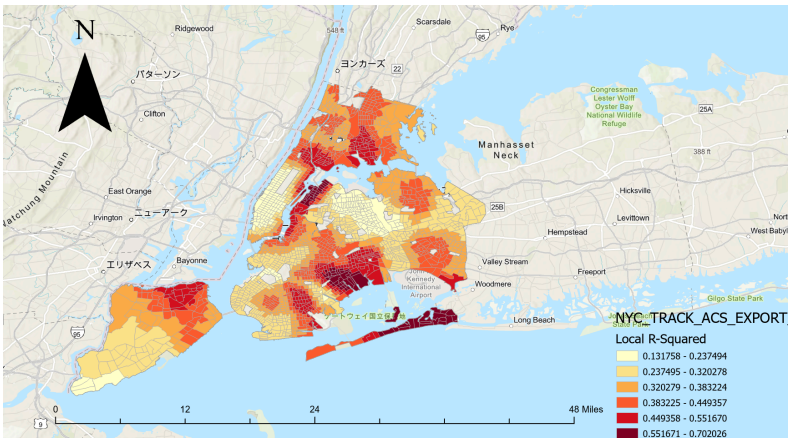
(i) Our OLS model had an R-squared of 0.3067. Our GWR model almost doubles this to 0.5717, and has the lowest of any AICc at -6776.3114. It is doing the best out of all of our models so far. It also shows almost no spatial autocorrelation, with Moran's I of GWR residuals being -0.001, with a p-value of 0.855.



A spatial autocorrelation report shows similarly good values, and no spatial autocorrelation. This is far better than the OLS and both the spatial lag and error models.

(iii) As mentioned previously, GWR runs a separate regression for each and every location and observation, using observations around it weighted by distance. Closer neighbours get more weight, which reflects that spatial relationships with variables can be different in different areas of NYC.

(iv)



Our choropleth map of local R-squared values reveals that the GWR model fits the data better in areas such as Staten Island and eastern Queens, where R-squared values reach as high as 0.70. In contrast, it performs worse in high-density areas such as Manhattan, where they fall as low as 0.13. This suggests that the relationship between the Gini index and its predictors is less spatially consistent in more urban environments.

4) Discussion

To conclude, this report examined the spatial patterns in income inequality (Gini index) in NYC using OLS, Spatial Lag Spatial Error, and GWR models. We found that OLS failed key assumptions for spatial regression, such as normality, homoscedasticity and independence. Whilst both the spatial lag and spatial error models were able to significantly improve fit and correct for these spatial mistakes, they did not solve them and still violated some key assumptions. GWR had the overall best performance, being able to explain over 57% of the variation with absolutely no residual spatial autocorrelation detected.

There were key limitations that remained for most of the models in this report. For example, homoscedasticity and autocorrelation remained key issues for spatial error and lag models. Whilst the GWR solved the spatial autocorrelation issue, it does not give p-values due to the previously mentioned issues with standard of error. Additionally, only a few predictors were included in this report, additional predictors could have further complicated results and resulted in the GWR performing poorly. Finally, this report did not include MGWR, which could have further improved issues surrounding assumptions and spatial autocorrelation.